problems/MHD/astroJetWuShu2018

DynSGS and positivity setup, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

Scope. Read off `user_inputs.jl` (with `lvl = 2`), the case files and the kernels they reach on `sm/MHD`. The case combines the conserved-form DynSGS with the *residual* sensor (for its Dirichlet-node treatment at the nozzle), element norms with fast-speed floors, no hold, the conservative element limiter, and flux-only guards.

1 The problem as set up

Deck key / item	Value as run	Source / remark
Equations	2D ideal GLM-MHD, state $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi)$	Wu & Shu (2018), Ex. 5.6
EOS	$p = (\gamma-1)(E - \frac{1}{2}\lvert\mathbf{m}\rvert^2/\rho - \frac{1}{2}\lvert\mathbf{B}\rvert^2 - \frac{1}{2}\psi^2)$, $\gamma = 1.4$	`pressure_mhd` with $\max(\rho, 10^{-14})$ in the kinetic term
Domain, mesh	right half $[0, 0.5] \times [0, 1.5]$, symmetry axis at $x = 0$; `AJ_half_20x60.msh` refined twice by p4est $\Rightarrow 80 \times 240$ quads ($\Delta^{elem} = 0.00625$)	`lvl = 2`
Discretization	CG spectral elements, LGL nodes, lumped mass; $N = 4$	
Time integration	`CarpenterKennedy2N54` (rebuilt with the limiter), fixed $\Delta t = 5 \times 10^{-7}/2^2 = 1.25 \times 10^{-7}$, $t_{end} = 2 \times 10^{-3}$	
Variables	`TOTAL()`, `"energy"`	
Initial data	ambient $\rho = 0.14$, $p = 1$, $\mathbf{v} = 0$, $\mathbf{B} = (0, \sqrt{200}, 0)$ ($\beta_a = 10^{-2}$)	`user_flux.jl:153-165`
Inflow	nozzle $\lvert x\rvert \le 0.05$ at $y = 0$: $\rho = 1.4$, $v = 800$ (Mach 800), $p = 1$; lip smoothed over one element (`aj_smooth` auto), inflow ramped on over $\tau = 2\Delta^{elem}_{\min}/800$ (`aj_tramp` auto)	`initialize.jl:144, 170`
Boundary conditions	nozzle: Dirichlet on all 9 slots; axis: $\rho u = 0$, $B_x = 0$; right/top: ambient state on all 9 slots; bottom outside the nozzle: $\psi = 0$ only	`user_bc.jl`
GLM cleaning	$c_h = \max(\lvert\mathbf{v}\rvert + c_f)$ over the initial *and the inflow* state, constant; $S_\psi = -(c_h/0.18)\psi$	
Sources	`:lsource => true` (ψ damping only)	
Other switches	`:lfilter`, `:lkep`, `:entropy_variables`, `:lrichardson`, `:ladapt` false; `:lschlieren => true`	

2 DynSGS: deck keys and effective values

Deck key / item	Value as run	Source / remark
`:visc_model`	`DSGS_MHD()`, `:lvisc => true`	`SGS.jl:1598`
`:`μ	all 1	
`:dsgs_sensor`	`"residual"`	element residual; Dirichlet nodes zeroed
`:dsgs_CR`, `:dsgs_Cmax`, `:dsgs_Cmin`	1.0, 0.5, 0	
`:dsgs_cutoff`	0	
`:dsgs_hold_steps`	**0**	ν from the first step
`:dsgs_norms`, `:dsgs_local_rel`	`"element"`, 1.0 (default)	
`:dsgs_rel`	1.0	unused with element norms
`:dsgs_fast_floors`	**true**	$c_f/c \approx 12$ in the ambient
`:dsgs_gamma`, `:dsgs_Prt`	$= \gamma_{mhd} = 1.4$, 1.0	
`:dsgs_conserved`	true	
`:dsgs_ref_weight`, `:dsgs_nazarov_energy`, `:dsgs_thermal_residual`	false	
`:ldsgs_nodal`, `:dsgs_freeze_stage`, `:dsgs_reference`, `:dsgs_nodal_rho`	false	

3 DynSGS as it runs for this case

$\Delta_K = \Delta_K^{elem}/(N+1) = 0.00125$, $h = \Delta t$.

1. **History, no hold.** Rotation once per step; with $H = 0$ the residual is live from the first RHS call, on a history seeded with the initial condition (first stage of step 1: BDF2 of (q^0, q^0, q^0), $D_t q = 0$).
2. **Element residual** (residual sensor, stage-consistent stencil): $R^s_{K,I} = |D_t q^s_I - r_I^{K,s}/m_I^K|$, $D_t q$ = BDF2 on (q^n, q^{n-1}, q^{n-2}) at $\tau = 0$, the three-point derivative through $(q(\tau), q^n, q^{n-1})$ at $\tau > 0$; r^K the element's own weak inviscid RHS.
3. **Dirichlet nodes.** Every node on a non-periodic boundary edge (nozzle and the rest of the bottom, axis, right, top) is listed by `_dsgs_boundary_pairs!`, once per element containing it; its entries are replaced by $m_I^K D_t q_I$, so $R \equiv 0$ there in all nine slots. Inflow shocks are sensed by the interior nodes of the same elements.
4. **Element normalization with fast floors** over the 25 nodes of K: $n^s_K = \max(S^s_K, \sigma^s_K) + 10^{-16}$, $\sigma_K = (\bar\rho_K,\ \bar\rho_K \bar c_{f,K},\ \gamma\bar p_K + |\bar{\mathbf{B}}_K|^2,\ \sqrt{\gamma\bar p_K + |\bar{\mathbf{B}}_K|^2})$ for $(\rho, \mathbf{m}, E, \mathbf{B})$, $\bar c_{f,K} = \sqrt{(\gamma\bar p_K + |\bar{\mathbf{B}}_K|^2)/\bar\rho_K}$.
5. **Ratio, cap, coefficient.** $\mathcal{R}_K = \max_{I\in K} \max_{s\le 8} R^s_{K,I}/n^s_K$, $w_K = \max_I(|\mathbf{v}_I| + \sqrt{\max(\gamma p_I/\rho_I + |\mathbf{B}_I|^2/\rho_I, 0)})$, $\nu_K = \max(0, \min(0.5\,\Delta_K w_K,\ \Delta_K^2 \mathcal{R}_K))$, no floor, no cutoff, one value per element, recomputed at every stage; ν_K on all nine slots.

4 The viscous operator

`user_primitives.jl` returns $\mathbf{q}$ itself: $\partial_t \mathbf{q} + \nabla\cdot\mathbf{F} = \nabla\cdot(\nu_K \nabla\mathbf{q})$ on $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi)$ — a Laplacian of the conserved state, mass diffusion on, no $\tau\cdot\mathbf{v}$ or resistive-work term.

5 Positivity

Conservative element limiter

Deck key / item	Value as run	Source / remark
`:lpositivity`	**true**	
`:positivity_method`	`"conservative"`	
`:positivity_rho_min`	1.4×10^{-7} ($= 10^{-6}\times$ ambient ρ)	
`:positivity_p_min`	10^{-6} ($= 10^{-6}\times$ ambient p)	
`:positivity_report`, `_every`	true, 250	250 limiter calls = 50 steps

How the limiter is hooked in

- `:lpositivity => true` and `:positivity_method => "conservative"` make `mod_inputs.jl:836-838` rebuild the deck's `CarpenterKennedy2N54` with the limiter as its stage limiter and `williamson_condition = false`:

  ```
  CarpenterKennedy2N54(stage_limiter!  = positivity_conservative_limiter!, williamson_condition = false)
  ```

  (`positivity_driver.jl:263`); any other solver, or one that already has a stage limiter, is refused.
- The limiter therefore runs before the right-hand side of stages 2–5 and once after stage 5: five times per step, on every rank. The state entering stage 1 is the limited end of the previous step; the initial condition itself is not limited.
- On the first call `_cpos_validate` (`positivity_driver.jl:271`) requires: the nine-field `qvars` $(\rho, \rho u, \rho v, \rho E, \rho w, B_x, B_y, B_z, \psi)$, 2D, continuous Galerkin, a lumped (vector) mass matrix, `TOTAL()`, CPU, no `:ladapt`, `:dsgs_gamma` set, both floors > 0. γ is taken from `:dsgs_gamma`, not from `PhysicalConst`.
- The node-wise *repair* of `apply_positivity!` (`rhs.jl:821`) returns at once in this mode: nothing is done inside `rhs!`.

What one call does

Let $\tilde{E} = E - \frac{1}{2}|\mathbf{B}|^2 - \frac{1}{2}\psi^2$ (the part of E the limiter may touch), $p(\rho, \mathbf{m}, \tilde{E}) = (\gamma-1)(\tilde{E} - |\mathbf{m}|^2/(2\rho))$, $\varepsilon_\rho = 1.4\times10^{-7}$, $\varepsilon_p = 10^{-6}$, $s = 64$, and $\mathrm{ulp}(x)$ the double-precision spacing at x.

1. **Mark** (`cpos_mark_bad!`): node I is bad unless $\rho_I \geq \varepsilon_\rho$ and $p_I \geq \varepsilon_p$ (a NaN node is therefore bad). The number of bad nodes is `MPI.Allreduce`d; if it is 0 everywhere the call returns and the state is untouched.
2. **Element pass** (`cpos_element_pass!`), for every element K holding a bad node, with the element mass weights $w_I = \omega_i \omega_j J_{K,ij}$, $W = \sum_I w_I$:
   - means $\bar{\rho} = \sum w_I \rho_I / W$, $\bar{\mathbf{m}}$, $\bar{E}$ (of $\tilde{E}$); round-off margins $\delta_\rho = s\,\mathrm{ulp}(\max(\max_I|\rho_I|, |\bar{\rho}|))$, $\bar{\rho}_g = \bar{\rho} - \delta_\rho$, $\varepsilon_\rho^+ = \varepsilon_\rho + \delta_\rho$, $\varepsilon_p^+ = \varepsilon_p + s(\gamma-1)\,\mathrm{ulp}(\max_I(|E_I| + \frac{1}{2}|\mathbf{B}_I|^2 + \frac{1}{2}\psi_I^2))$;
   - if the mean is not admissible ($\bar{\rho}_g \leq \varepsilon_\rho^+$ or $p(\bar{\rho}_g, \bar{\mathbf{m}}, \bar{E}) \leq \varepsilon_p^+$, or not finite) the element is counted as *failed* and skipped;
   - density: if $\min_I \rho_I < \varepsilon_\rho^+$, $\theta_\rho = \mathrm{clamp}\big((\bar{\rho} - \varepsilon_\rho^+)/(\bar{\rho} - \min_I \rho_I), 0, 1\big)$, else $\theta_\rho = 1$; $\tilde{\rho}_I = \bar{\rho} + \theta_\rho(\rho_I - \bar{\rho})$;
   - pressure: for every node with $p(\tilde{\rho}_I - \mathrm{margin}, \mathbf{m}_I, \tilde{E}_I) < \varepsilon_p^+$, $t_I \in [0,1]$ is the smallest root of the quadratic $(\bar{\rho}_g + t\,d\rho)(\bar{E} - e^+ + t\,dE) - \frac{1}{2}|\bar{\mathbf{m}} + t\,d\mathbf{m}|^2 = 0$, $e^+ = \varepsilon_p^+/(\gamma-1)$, $d\rho = \tilde{\rho}_{I,g} - \bar{\rho}_g$, $d\mathbf{m} = \mathbf{m}_I - \bar{\mathbf{m}}$, $dE = \tilde{E}_I - \bar{E}$ (computed with the cancellation-free root formula); $\theta_p = \min_I t_I$ ($\theta_p = 0$ if some $\tilde{\rho}_{I,g} \leq 0$);
   - increments, accumulated as $w_I\,\delta q_I$:

     $$\delta\rho_I = (\theta_p\theta_\rho - 1)(\rho_I - \bar{\rho}), \quad \delta\mathbf{m}_I = (\theta_p - 1)(\mathbf{m}_I - \bar{\mathbf{m}}), \quad \delta E_I = (\theta_p - 1)(\tilde{E}_I - \bar{E}), \quad \delta\mathbf{B}_I = \delta\psi_I = 0.$$

     Since $\bar{q}$ is the w-weighted mean, $\sum_I w_I \delta q_I = 0$ in every element: ρ, $\rho\mathbf{v}$ and E are conserved exactly; $\mathbf{B}$ and ψ (hence $\nabla\cdot\mathbf{B}$) are never touched.
3. **Assemble and apply** (`DSS_global_RHS!`, `cpos_add!`): the element increments are summed over elements, ranks and periodic copies and divided by the lumped global mass, $q_I \leftarrow q_I + M_I^{-1}\sum_K w_I^K \delta q_I^K$. At a node shared by several elements the update is the mass-weighted average of their increments, so the assembled node need not reproduce the element's limited value exactly.

4. **Node floor** (`cpos_floor!`, not conservative), for the finite nodes still out of bounds (e.g. in failed elements): $\rho \leftarrow \varepsilon_\rho + s\,\mathrm{ulp}(\varepsilon_\rho)$ if $\rho < \varepsilon_\rho$ (mass added); then with $k = |\mathbf{m}|^2/(2\rho)$ and $e^+ = \varepsilon_p/(\gamma-1) + s\,\mathrm{ulp}(|E| + \frac{1}{2}|\mathbf{B}|^2 + \frac{1}{2}\psi^2)$, if $\tilde{E} - k < e^+$: either scale $\mathbf{m}$ by $\sqrt{(\tilde{E} - e^+)/k}$ (E unchanged) when $\tilde{E} > e^+$, or set $\mathbf{m} = 0$ and raise E by $e^+ - \tilde{E}$ (energy added).
5. **Report** (`_cpos_report`): every 250 limiter calls (`:positivity_report_every`), when the global counts have grown: element limitings, $\min\theta_\rho$, $\min\theta_p$, failed elements, node floors, and in how many calls the limiter was active.

Flux-only guards (`user_flux.jl:51-58`)

Inside `user_flux!` only: $\rho_g = \max(\rho, 10^{-14})$ in the velocities and in the kinetic energy of `pressure_mhd`, and $p \leftarrow \max(p, 10^{-6})$ (`p_floor_mhd`, equal to ε_p). They change the flux of a node that is momentarily inadmissible, not the state; with the limiter active every node the flux sees already satisfies $\rho \geq \varepsilon_\rho$, $p \geq \varepsilon_p$ up to the round-off margins above.

6 Observations

1. Deck and comments agree with the code. The `"residual"` sensor is chosen explicitly for its boundary-node treatment, which the `"legacy"` path does not have.
2. p in this beam is $\approx 5.6 \times 10^{-6}$ of E; the deck comment on `:dsgs_norms` explains the element norms by that ratio, and `Positivity.jl:275-289` quantifies how small $p_{\min}$ may usefully be ($(\gamma-1)\,\mathrm{ulp}(E) \approx 2.3 \times 10^{-11}$ against $p_{\min} = 10^{-6}$).

Appendix: code map (branch `sm/MHD`, commit `dcc3653`)

Mechanism	Where (file:line)
Deck defaults of every `:dsgs_*` key	`src/io/mod_inputs.jl:1085-1376`
Deck defaults of `:lpositivity`, floors, report	`src/io/mod_inputs.jl:456-477`
`:positivity_method`; rebuild of the RK solver	`src/io/mod_inputs.jl:828-838`
DynSGS history rotation, stage weights, hold, freeze	`src/kernel/operators/rhs.jl:927-984`
Stencil selection `_dsgs_stencil`	`rhs.jl:141`
Hold length `_dsgs_hold_steps` ($\max(2, H) + 1$ rotations)	`rhs.jl:182`
Residual right-hand side (element / legacy / Dirichlet)	`rhs.jl:188-280`; legacy fill `rhs.jl:282`; boundary zeroing `rhs.jl:313, 346`
1D `DSGS_MHD` dispatch	`rhs.jl:1398-1470`
2D `DSGS_MHD` dispatch (sets `dsgs_ref_weight`, `dsgs_split_energy`, `dsgs_nodal_rho`)	`rhs.jl:1517-1610`
2D viscous assembly `_viscous_rhs_el_2d_dsgs!`	`rhs.jl:1796`
2D weak viscous flux `_expansion_visc!` (conserved flag, ref. weight, split energy, $\tau \cdot \mathbf{v}$, resistive work)	`rhs.jl:2943-3210` (`lcons` 2959, weight 2978, split 3050, $\tau \cdot \mathbf{v}$ 3076, $\eta\,\mathbf{B} \cdot \nabla\mathbf{B}$ 3091)
1D weak viscous flux	`rhs.jl:2823`; element loop `rhs.jl:1321`
2D `DSGS_MHD` element kernel	`src/kernel/physics/SGS.jl:1598-1912`
1D `DSGS_MHD` element kernel	`SGS.jl:1929-2120`
Nodal kernels (off in every MHD deck; BW can switch it on with `JEXPRESSO_BW_NODAL=1`)	`SGS.jl:2144` (1D), `SGS.jl:2515` (2D)
Normalization floor `_dsgs_denom`, `:dsgs_rel` multiplier	`SGS.jl:676-679`; set per call in `rhs.jl:192`
Smoothness cutoff `_dsgs_cut`	`SGS.jl:703`
`SGS_diffusion(::DSGS_MHD)` (nodal-ρ switch)	`SGS.jl:547-569`
Element length Δ_K^{elem} (2D: min of corner extents)	`src/kernel/mesh/mesh.jl:6430-6457`; 1D Δx: `mesh.jl:5863`
`uprimitive` width ($n_{eqs} + 2$ in 2D, $n_{eqs} + 1$ in 1D)	`src/kernel/globalStructs.jl:139-153`
Realizability repair (`"repair"`, unused by the MHD decks)	`src/kernel/positivity/Positivity.jl:294`; driver `positivity_driver.jl:124`; called in `rhs.jl:821`

Mechanism	Where (file:line)
Conservative element limiter (`"conservative"`)	`Positivity.jl:433-559`; driver `positivity_driver.jl:245-337`
Main `solve` call (`saveat` non-empty ⇒ `calck = true`)	`src/kernel/solvers/TimeIntegrators.jl:855`

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)